

CONCEPTION TO EMBRYO

THE EMBRYONIC CELLS REPEATEDLY DIVIDE, AND BECOME IMPLANTED INTO THE UTERUS LINING.

The first eight weeks in the uterus are known as the embryo stage, in which the fertilized egg becomes a tiny human body, no larger than a thumb. The fertilized egg develops into an enlarging cluster of cells, the blastocyst. Some cells will form the baby's body; while others become the protective membranes or the placenta, which nourishes the embryo and removes waste products.

Fallopian tube

Conveys zygote towards uterus

First cleavage

Large zygote splits itself into two cells

Cilia

Microhairs waft the zygote along

Goblet cells

Secrete fluid into tube

2 ZYGOTE

The fertilized egg passes along the fallopian tube. Within 24–36 hours it has divided into two cells, then 12 hours later into four cells, and so on. This process is known as cleavage. At each stage the resulting cells become smaller, gradually approaching normal body cell size.

1 FERTILIZATION

Fertilization takes place in the fallopian tube, when the head of the sperm cell, or spermatozoon, penetrates the much larger ripe egg cell, or mature ovum. This forms a single cell – the fertilized egg, or zygote, which contains 23 pairs of chromosomes (see p.286).

Morula

Fallopian tube lining

Cilia

3 MORULA

A cluster of 16–32 cells, the morula leaves the fallopian tube and enters the uterus at around 3–4 days after fertilization.

Fallopian tube

Fimbriae

Ovary

Ovarian ligament

Ovum (egg cell)

Up to 0.1mm ($\frac{1}{250}$ in) across (huge compared to other cells); contains 23 maternal chromosomes

Corona cell

Secretes chemicals to aid egg development

Tail of sperm

Lashes to propel sperm towards egg

Sperm head

Contains 23 paternal chromosomes

Acrosome

"Cap" of sperm head, which penetrates egg cell membrane